When Names Mislead: Semantic-Ownership Susceptibility in LLM-Based Root Cause Analysis

ANONYMOUS AUTHOR(S)

Natural software names provide useful priors to a diagnostic model, but the same priors can mislead when attached to the wrong component. Ordinary accuracy and anonymization tests report performance effects while leaving the destination of a misleading cue unobserved. We introduce a semantic-ownership stress test that reassigns natural labels within entity type while keeping diagnostic evidence, canonical IDs, and candidate availability fixed. A recorded source–destination map identifies the same-type wrong entity receiving the true root's label. We evaluate two model configurations on 87 RCA100 incidents, using three preregistered assignments with and without explicit topology. With topology visible, Qwen selected this destination 45/261 times versus 1/261 under a typed-opaque control, an attraction contrast of 16.86 percentage points (95% CI [11.49, 22.61]) with six strict stable switches. The pattern replicated without topology at 12.26 points ([7.66, 17.62]) with four strict switches. A descriptive natural-name reference yields similar contrasts, and the shifts include both correct-to-wrong and wrong-to-another-wrong transitions. DeepSeek showed smaller positive contrasts of 2.30 and 1.92 points and no strict switch. Meaningful names can aid diagnosis yet become a measurable source of destination-specific error when their ownership is wrong; susceptibility varied across the two model configurations.

CCS Concepts: • **Software and its engineering** → **Software reliability**; *Software testing and debugging*; • **Computing methodologies** → *Natural language processing*.

Additional Key Words and Phrases: root-cause analysis, large language models, semantic shortcuts, identifier robustness, behavioral testing

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1 Introduction

A production incident presents a diagnostic model with two distinct information sources. Telemetry records what happened: an error propagated, a latency distribution shifted, or a pod restarted. Software names communicate what components are expected to do: `paymentservice`, `checkout`, and `postgres` carry strong role semantics before any trace is read. These cues are useful in normal operation, yet stale aliases, copied configuration, service repurposing, and inconsistent instrumentation can make them conflict with runtime evidence. A reliable root-cause decision should not move to an evidence-inconsistent component merely because that component receives a plausible name.

Ordinary accuracy reports whether a diagnosis is correct, and natural-versus-anonymized comparisons reveal whether names affect performance. Identifying the destination of a misleading cue requires a directional test. Anonymization removes useful information and changes the input distribution, so an accuracy decrease can reflect lost context as well as shortcut following. The sharper question is directional: when the semantic label associated with the true root is reassigned to a same-type wrong entity, does the diagnosis move toward that destination while diagnostic evidence and candidate availability stay fixed?

Prior software-engineering studies show that identifiers influence learned code models, that familiar but incorrect names can dominate structural cues, and that controlled renaming is a useful robustness test [2, 6, 10, 14]. We build on that foundation by defining *semantic-ownership susceptibility*: excess selection of the specific wrong entity that inherits a root-associated label,

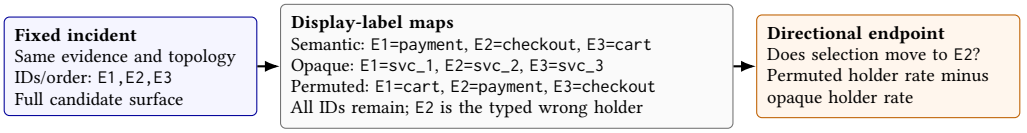


Fig. 1. Integrity-preserving semantic-ownership intervention. Across label conditions within a fixed topology setting, stable IDs, evidence, and candidate membership and order are fixed. The semantic-to-permuted contrast additionally preserves the complete within-type natural-label multiset while moving the root-owned label to a provenance-defined same-type wrong destination.

relative to that entity's attraction under a candidate-specific typed-opaque control. This destination-specific construct differs from both generic identifier sensitivity and *semantic usefulness*, the ordinary accuracy benefit of meaningful names.

To make that directional question testable, every condition must show the same entities with the same stable IDs and evidence. Names must move only within entity type, and the experiment must record which destination received each source name. These controls separate a move toward the designated entity from a changed choice set or an ambiguous string match. Figure 1 shows the intervention.

We apply this instrument to 87 incidents, using three preregistered assignments, two fixed model configurations, and contexts with and without explicit topology. Qwen's typed-holder attraction is 16.86 pp (95% CI [11.49, 22.61]) with topology and 12.26 pp ([7.66, 17.62]) without it. The corresponding DeepSeek contrasts are 2.30 pp and 1.92 pp, with no strict switch. Of the 13 incidents Qwen diagnoses correctly under semantic labels with topology, 10 move to the designated holder under at least one of the three assignments; other shifts redirect an existing error. The destination-specific analysis therefore adds information beyond aggregate accuracy: the shift is substantial for Qwen, smaller for DeepSeek, and includes redirection of existing errors.

This paper makes three contributions:

- **Directional measurement.** We formulate semantic-ownership susceptibility as selection of the specific wrong entity that receives a root-associated name, separating destination-specific redirection from generic identifier sensitivity and semantic usefulness.
- **Controlled intervention.** We reassign complete within-type name maps while preserving evidence, canonical identity, candidate membership and order, and we derive the destination from recorded source–destination provenance.
- **Behavioral evidence.** We quantify attraction, natural-name and opaque references, diagnostic starting states, and incident coverage. Qwen shows substantial attraction in both topology contexts; DeepSeek shows smaller positive averages without a strict stable switch.

2 Background and Related Work

2.1 Names as Software Semantic Cues

Identifiers are a major carrier of human-readable semantics in software and part of the statistical regularities learned by code models [1]. Gao et al. show that identifiers can support or mislead neural code comprehension and use counterfactual inference to reduce harmful effects [6]. Miceli-Barone et al. swap Python built-ins while preserving executable bindings and find that language models often prefer familiar but incorrect meanings [10]. Renaming benchmarks likewise reveal substantial robustness failures across pretrained code models [14].

Recent studies contrast structural and lexical cues directly. Budiman and Minku find name-dominated attributions in design-pattern detection, with anonymization only partly closing the

robustness gap [2]. Rafi et al. study order effects in LLM fault localization and use renamed methods to examine a competing leakage explanation [11]. These studies measure accuracy, attribution, or general robustness under renaming. We instead evaluate a relational outcome: selection of a wrong destination specified in advance by the name map.

2.2 Shortcut Learning and Behavioral Testing

Shortcut learning describes predictive behavior that exploits features correlated with the target while bypassing the evidence intended by the task [7]. Language models exhibit lexical and prompt-level shortcuts under controlled distribution shifts [5, 13]. We use *susceptibility* as a directional behavioral property: the output moves toward a typed destination under a controlled label reassignment.

Behavioral testing provides the methodological template. CheckList separates ordinary capability tests, invariance tests, and directional expectation tests [12]. Semantic versus opaque accuracy is a capability contrast. Ownership reassignment is directional: a wrong candidate should not become more attractive merely because it receives the root-associated label. The opaque holder rate provides a candidate-specific label-neutral reference, and strict stable switches identify cases where semantic and opaque predictions agree before permutation redirects the verdict.

2.3 LLM-Based Root-Cause Analysis

Microservice RCA combines metrics, logs, traces, events, and dependency relations across services and infrastructure. Systems such as MicroHECL and Eadro learn from topology and multi-source telemetry [8, 9]. Recent work explores LLM agents for incident diagnosis and resolution, including multi-agent coordination and autonomous failure analysis [4, 15]. Natural component descriptions can help these systems, but they also make aggregate accuracy an ambiguous measure of evidence grounding.

RCA100 provides multimodal telemetry, stable entity registries, and per-incident topology for chaos-injected incidents [3]. These properties let us separate canonical identity from display labels and hold the diagnostic evidence fixed. Our focus is black-box diagnostic behavior under controlled naming interventions.

3 Semantic-Ownership Stress Testing

3.1 Research Question

Our central research question is:

> *When a semantic label associated with the true root is reassigned to a same-type wrong entity, does an LLM-based RCA system redirect its diagnosis toward that destination under otherwise unchanged diagnostic evidence and candidate availability?*

We separate two subordinate questions. **SQ1 (semantic usefulness)** asks whether meaningful labels improve root accuracy relative to typed opaque labels. **SQ2 (semantic-ownership susceptibility)** asks whether reassignment increases selection of the typed wrong holder relative to that same candidate under opaque labels. The two questions target distinct behaviors: SQ1 quantifies the accuracy benefit of meaningful context, whereas SQ2 measures redirection when that context becomes misleading.

3.2 Separating Identity, Evidence, and Names

For incident i, each candidate $e \in \mathcal{E}_i$ has a stable canonical ID, entity type $\tau(e)$, and semantic display label $L_i(e)$. Outputs always use canonical IDs. The semantic condition displays L_i. The opaque condition replaces each label with a deterministic typed token that preserves coarse ontology while suppressing role semantics.

3.3 Reassigning Names without Changing Candidates

For each preregistered seed s, the intervention reassigns names among entities of the same type. Types lacking a nontrivial derangement use identity carry: their entities stay visible and retain their names, preserving the candidate list. Within a fixed topology setting, every label condition therefore preserves candidate membership and order, stable IDs, evidence, and nonlabel prompt fields. Semantic and permuted conditions also share the complete within-type natural-label multiset, whereas opaque conditions replace natural names with deterministic typed tokens. Formally, the generator records inverse bijections `source_by_destination` and `destination_by_source`. Intervention construction uses only public case structure and the seed; restricted roots enter during evaluation after responses are fixed.

Table 1. Condition matrix. Every cell retains the full candidate surface; only display labels and the prespecified topology context vary.

Condition	Labels	Candidates	Evidence	IDs/order	Topology
Semantic/on	original	full	fixed	fixed	fixed
Opaque/on	typed opaque	full	fixed	fixed	fixed
Permuted/on	within-type full map	full	fixed	fixed	fixed
Semantic/off	original	full	fixed	fixed	absent
Opaque/off	typed opaque	full	fixed	fixed	absent
Permuted/off	within-type full map	full	fixed	fixed	absent

3.4 Tracking the Destination that Receives the Root Name

Restricted roots are loaded only after requests and responses are fixed. For gold root r_i, the seed-specific typed holder is

$$\text{holder}(i,s) = d \quad \text{such that} \quad \text{source_by_destination}_{i,s}(d) = r_i, \quad \tau(d) = \tau(r_i), \quad d \neq r_i. \tag{1}$$

In plain terms, the holder is the same-type wrong entity that receives the root's original name. The evaluator resolves it directly from the recorded map, avoiding display-string lookup. Every analyzed case–seed row has one observable typed wrong holder. The public generator remains root-independent; only after responses are fixed does the restricted evaluator join the root to the already recorded map.

3.5 Estimands and Decision Rules

Semantic usefulness for model m and topology context t is the paired incident mean

$$\Delta^{m,t}_{\text{sem}} = \frac{1}{N} \sum_i \left[I(\hat{r}^{m,t}_{i,\text{sem}} = r_i) - I(\hat{r}^{m,t}_{i,\text{opaque}} = r_i) \right]. \tag{2}$$

For holder attraction, let $A^{i,s,m,t}_{\text{perm}}$ indicate selection of holder(i,s) under permutation and $A^{i,s,m,t}_{\text{opaque}}$ selection of the same candidate under the opaque condition. The case-clustered estimand is

$$\Delta^{m,t}_{\text{attr}} = \frac{1}{N} \sum_i \frac{1}{3} \sum_s \left(A^{i,s,m,t}_{\text{perm}} - A^{i,s,m,t}_{\text{opaque}} \right). \tag{3}$$

A *new switch* occurs when neither semantic nor opaque prediction selects the holder and the permuted prediction does. A *strict stable switch* additionally requires identical semantic and opaque

predictions. For SQ1, model-specific semantic benefit requires $\Delta_{\text{sem}} > 0$ and a strictly positive lower 95% confidence bound; cross-model confirmation requires both evaluated configurations to meet that rule. For SQ2, the model-specific criterion requires $\Delta_{\text{attr}} > 0$, a strictly positive lower bound, and at least one strict stable switch, again with both configurations required for cross-model confirmation. Topology on is primary; topology off applies the same rules as internal replication. Topology moderation lies outside the design's estimand.

After the registered outcomes were known, we specified descriptive analyses of the same responses: a semantic-label reference for each assignment-specific destination, semantic starting states, incident coverage, and strict-switch opportunities. They report both case–assignment rows and distinct incidents and treat the three assignment rows as clustered within incident. The registered estimators, criteria, and decisions remain fixed.

4 Study Design

4.1 Dataset and Confirmatory Population

RCA100 v1.1 contains 103 chaos-drill incidents from a Kubernetes deployment of the OpenTelemetry demo store. Each incident provides metrics, logs, traces, events, alerts, topology, and a restricted answer key [3]. A deterministic split assigned 16 incidents to development and 87 to confirmation before the reported confirmatory outcomes were observed. The same 87 incidents were used in an earlier intervention, so researchers had prior exposure to this incident set; the development set remained separate. That earlier instrument changed candidate availability and used a display-string destination; Appendix C reports its measurement audit. All confirmatory roots resolve to one canonical entity, and every typed holder is present in the submitted full candidate surface.

Each condition derives from the same bounded *medium* diagnostic view. The compiler selects evidence before label intervention and assigns stable candidate and evidence IDs. Across the 87 incidents, a view contains 253–1,951 candidate entities (median 275), 51–67 evidence units (median 60) spanning alerts, events, logs, metrics, and traces, and 16–20 entity types (median 20). Topology-on views contain 32 nodes and 0–14 edges (median 4). Evidence units retain stable targets and provenance; names appear in the separate candidate map. Semantic, opaque, and permuted conditions share the parent evidence view, and topology-off variants remove topology consistently across label modes.

4.2 Conditions and Seeds

The study uses semantic, typed-opaque, and three permuted conditions under topology on and off. Seeds 20260901, 20260902, and 20260903 were prespecified before confirmation. Pairing uses the same incident and model configuration; the three permutation observations remain clustered within incident.

4.3 Models and Inference Settings

We evaluate two fixed model configurations: Qwen `qwen3.7-plus-2026-05-26` and DeepSeek `deepseek-v4-pro`. Both receive identical diagnostic instructions and the same four-field JSON contract: root ID, confidence, up to three evidence IDs, and a brief justification. Temperature is 0, top-p is 1, tools and streaming are disabled, model thinking is disabled, and the maximum output is 1024 tokens.

Both providers use provider-native structured-output enforcement. Their responses pass through the same local parser, contract validator, and prediction-validity rule.

4.4 Output Validity and Statistical Analysis

Output validity distinguishes root-prediction validity from supporting-evidence-reference conformance. A parseable response contributes a root prediction when its required structural fields are valid and its root is a supplied candidate or `UNKNOWN`. More than three evidence IDs or references outside the supplied evidence domain are reported separately because citation sufficiency lies outside the scientific endpoints. Malformed JSON, missing or out-of-candidate roots, empty primary content, and truncated primary generation are prediction-invalid and contribute zero positive events. Primary responses enter the analysis unchanged.

All intervals use 10,000 bootstrap resamples of the 87 incidents with seed 20260905. The three assignment rows move together when an incident is resampled. SQ1 also reports discordant-pair counts and exact two-sided McNemar tests, while the protocol-defined confidence interval remains its confirmation criterion. All confirmatory cases and assignments enter the analysis.

5 Results

5.1 Directional Attraction to the Name Recipient

With topology visible, Qwen selects the typed wrong holder 45 times across 261 permuted case–assignment rows and the same assignment-specific destinations once under the opaque control. The incident-clustered attraction contrast is 16.86 pp (95% CI [11.49, 22.61]). Forty-four selections are new switches, and six are strict stable switches. This cell meets the prespecified model-specific criterion. DeepSeek selects the holder seven times versus once, a smaller 2.30 pp contrast ([0.38, 4.98]); six are new switches, but none is strict. The primary cross-model criterion is therefore not met.

Figure 2 and Table 2 show both topology contexts. The Qwen pattern recurs without topology: 33 holder selections versus one opaque selection yield a 12.26 pp contrast ([7.66, 17.62]) with 32 new and four strict switches. DeepSeek selects the holder six times versus once, a 1.92 pp contrast ([0.00, 4.60]) with five new and no strict switch. Qwen thus meets the model-specific rule in both contexts, whereas DeepSeek's positive averages leave the complete rule unmet. The topology-off context serves as an internal replication.

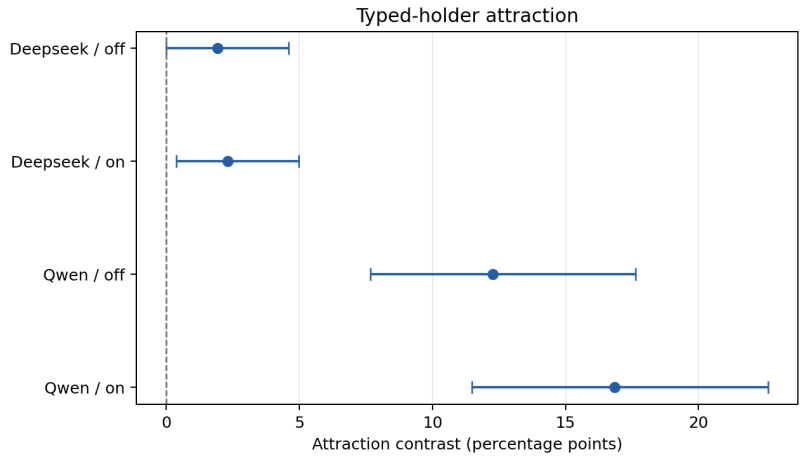


Fig. 2. Typed-holder attraction in percentage points. Points are permuted minus assignment-matched opaque holder-selection rates; intervals resample the 87 incidents while retaining their three assignments.

Table 2. Registered typed-holder attraction over 87 incidents and three preregistered assignments. Contrasts and intervals are percentage points. Confirmation requires a positive point estimate, positive lower confidence bound, and at least one strict stable switch.

Top.	Model	Perm.	Opaque	Δ_{attr}	95% CI	New	Strict
On	Qwen	45	1	16.86	[11.49,22.61]	44	6
On	DeepSeek	7	1	2.30	[0.38,4.98]	6	0
Off	Qwen	33	1	12.26	[7.66,17.62]	32	4
Off	DeepSeek	6	1	1.92	[0.00,4.60]	5	0

Across the three assignments, Qwen's topology-on contrasts are 12.64, 14.94, and 22.99 pp; topology-off contrasts are 9.20, 11.49, and 16.09 pp. DeepSeek's topology-on contrasts are each 2.30 pp, while topology-off contrasts are 2.30, 1.15, and 2.30 pp. The aggregate direction is consistent across assignments, while the incidents contributing to movement vary by assignment.

5.2 Natural-Name Reference and Diagnostic Transitions

Qwen's destination-specific shift also appears against the natural-name reference: 16.48 pp with topology and 11.88 pp without it. The corresponding DeepSeek contrasts are 2.30 pp and 1.92 pp. Table 3 aligns each assignment-specific destination with the shared semantic response. Semantic and permuted conditions preserve the same natural-name multiset. The treatment nevertheless reassigns the full map, so the comparison captures the joint naming context.

Table 3. Descriptive holder references and semantic starting states. Counts are case–assignment rows; Cases is the number of distinct incidents with any permuted holder selection.

Top.	Model	Perm.	Sem.	Opaq.	Correct→H	Other wrong→H	Already H	Cases
On	Qwen	45	2	1	17	27	1	30
On	DeepSeek	7	1	1	1	5	1	5
Off	Qwen	33	2	1	13	19	1	22
Off	DeepSeek	6	1	1	2	3	1	4

Every permuted holder selection begins from a valid, non-`UNKNOWN` semantic response. Of the 13 incidents Qwen diagnoses correctly under semantic labels with topology, 10 move to the holder under at least one of three assignments, accounting for 17 of 39 case–assignment rows. Without topology, the corresponding figures are eight of 14 incidents and 13 of 42 rows. The 27 and 19 other-wrong-to-holder rows cover 19 and 13 incidents, respectively. The six topology-on strict switches cover four incidents and the four topology-off strict switches cover three; in all ten rows, semantic and opaque controls select the same wrong entity whereas the permuted condition selects the designated name recipient. Strict switches measure directional redirection; accuracy-loss transitions are reported separately.

DeepSeek had ample rows on which a strict switch could occur: semantic and opaque predictions were valid and identical and selected another candidate in 140 topology-on and 143 topology-off case–assignment rows, covering 47 and 48 incidents. These rows provide ample stable-control opportunities, yet no strict switch occurred. Its small positive average contrasts and several correct- or other-wrong-to-holder transitions still indicate residual sensitivity, leaving robustness unresolved.

5.3 Semantic Usefulness

Table 4 reports the separate capability contrast. With topology, Qwen identifies 13 roots under semantic labels and one under opaque labels, a 13.79 pp difference (95% CI [6.90, 21.84]). DeepSeek identifies four versus one, a 3.45 pp difference whose lower bound is zero. The primary cross-model semantic-benefit criterion is therefore not met. Without topology, Qwen identifies 14 roots versus two and DeepSeek five versus one; both intervals have positive lower bounds. For DeepSeek without topology, all four discordant incidents favor semantic labels; consequently, the protocol-defined percentile-bootstrap interval excludes zero, whereas the discrete exact McNemar test yields $p = .125$. Meaningful names can therefore provide useful context even where incorrect ownership creates a directed liability.

Table 4. Semantic usefulness over 87 incidents. Differences and intervals are percentage points; Disc. S/O gives semantic-only/opaque-only correct pairs, and McNemar p values are exact and two-sided. Prediction-invalid and UNKNOWN outputs count as incorrect.

Top.	Model	Semantic	Opaque	Difference	95% CI	Disc. S/O	McNemar p
On	Qwen	13/87	1/87	13.79	[6.90,21.84]	12/0	0.00049
On	DeepSeek	4/87	1/87	3.45	[0.00,8.05]	3/0	0.250
Off	Qwen	14/87	2/87	13.79	[5.75,21.84]	13/1	0.00183
Off	DeepSeek	5/87	1/87	4.60	[1.15,9.20]	4/0	0.125

6 Discussion

6.1 Useful Context Can Become a Directed Liability

Semantic usefulness and ownership susceptibility answer different reliability questions. Natural names improve Qwen's root accuracy relative to typed opaque labels, yet the same configuration follows a root-associated name toward a wrong recipient when ownership is changed. The opaque reference asks whether the destination becomes attractive relative to a label-neutral presentation; the semantic reference asks whether it becomes attractive relative to its own natural name while the natural-name multiset is preserved. Their similar Qwen contrasts strengthen the destination-specific interpretation.

DeepSeek shows small average shifts under both reference analyses and no strict stable switch, separating average attraction from the stronger agreement-based transition captured by the strict-switch rule.

6.2 The Shift Includes Two Kinds of Diagnostic Error

The transition decomposition separates practical harm from directional organization of existing mistakes. Some permuted holder selections replace a correct semantic diagnosis: 17 Qwen topology-on rows across 10 incidents and 13 topology-off rows across eight. More often, the semantic prediction is already another wrong entity before the permuted prediction moves to the designated holder. Those wrong-to-another-wrong transitions preserve the total error count while revealing how incorrect name ownership structures where errors land.

Strict stable switches make a narrower claim. In all ten Qwen rows, semantic and opaque controls agree on the same wrong entity before permutation redirects the verdict. They provide controlled behavioral witnesses of directional redirection; accuracy-loss counts and repeated-run stability are separate quantities. Reporting both row and incident counts is essential because three assignments from one incident are correlated.

6.3 Implications for RCA Evaluation

Accuracy under natural names can mix telemetry interpretation with useful semantic priors. Anonymization estimates the cost of removing those priors, whereas ownership tests identify the destination of a misleading cue. RCA evaluations should therefore report ordinary semantic performance and separately introduce controlled name-ownership conflicts. Such tests need stable machine IDs, unchanged candidate availability, source–destination provenance, and incident-clustered analysis.

Operational interfaces can support these tests by binding evidence and selected roots to canonical IDs while treating readable names as auditable attributes. Release-time evaluation can reassign complete typed name maps and verify that evidence, candidates, and nonlabel fields remain unchanged. The present evidence supports this practice as an audit mechanism; evaluating stable IDs as a mitigation remains future work.

7 Threats to Validity

Construct validity. The intervention reassigns a full joint label map, so its estimand concerns the combined naming context. Typed opaque labels retain entity-type information, and holder attraction is a behavioral endpoint. The natural-name holder reference and transition decomposition are post hoc descriptive analyses specified after the registered outcomes were known; the registered confirmatory endpoints remain unchanged.

Internal validity. Researchers had prior exposure to the same 87 incidents through the earlier intervention. The current maps, estimator, and decision rules were fixed before the reported responses, but this history limits claims of a wholly independent replication. The two configurations are remote provider models; fixed identifiers, deterministic decoding, and paired conditions constrain execution, but provider-side nondeterminism and unobserved updates can remain.

Output validity. A small fraction of confirmatory outputs are prediction-invalid, and some otherwise valid root predictions have auxiliary supporting-evidence-reference violations. Prediction-invalid outputs count as incorrect or nonselections, whereas valid roots remain in the analysis when only citation constraints fail because citation sufficiency lies outside the scientific endpoints. Different validity conventions could alter response rates. Under the adopted coding rule, invalid outputs contribute zero positive holder events.

Statistical conclusion validity. Inference resamples 87 base incidents and clusters the three assignments within each incident. The strict-switch rule yields sparse event counts, especially for DeepSeek. The descriptive semantic reference reuses one response across three assignment-specific destinations, so incident-clustered summaries remain essential. The topology-off comparison serves as internal replication, with topology moderation outside the estimand.

External validity. The evaluated configurations have limited absolute root accuracy. The reported contrasts therefore characterize destination-specific susceptibility in these configurations. The evaluation covers one chaos-injected microservice benchmark and two model configurations; operational naming drift may differ from a joint within-type map. Broader claims require replication across benchmarks, model configurations, and natural naming failures.

8 Conclusion

We tested whether moving a root-associated name to a same-type wrong entity redirects LLM-based RCA toward that destination while evidence, stable IDs, and candidate availability remain fixed. Qwen shows substantial attraction in both topology contexts; DeepSeek shows smaller positive

averages without a strict stable switch. The Qwen shifts include both correct diagnoses displaced to the wrong recipient and existing errors redirected toward it.

Names can therefore be useful context and a destination-specific liability when their ownership is wrong. Controlled ownership tests add a directional question to RCA evaluation: where does a diagnosis move when naming ownership changes? This question complements ordinary accuracy and anonymization by identifying the particular wrong entity that inherits a trusted name.

Data Availability

An anonymized review package is available online. It includes the frozen protocol and analysis plan, condition manifest, intervention-integrity summaries, aggregate confirmatory and descriptive results, the public full-surface permutation implementation, the restricted typed-holder evaluator, and review-only tables for offline endpoint inspection. The review-only component contains source-resolved roots, typed-holder mappings, and case-level predictions; it remains isolated from construction-time assets and was never provided to the evaluated models. Licensed raw telemetry, full prompts and payloads, raw model responses, provider logs, and credentials are excluded. RCA100 v1.1 is available from the maintainers' AgenticOpsEval repository under CC BY-NC-SA 4.0 at https://www.aiops.cn/gitlab/aiops-live-benchmark/agenticopseval. Upon acceptance, we will publicly release all redistributable replication materials while preserving the dataset's license and the separation of restricted evaluation assets.

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A Detailed Condition and Execution Protocol

A.1 Condition matrix

Each provider receives ten conditions per incident: semantic and opaque controls plus three full-surface permutations, each evaluated with and without topology. The permutations use seeds 20260901, 20260902, and 20260903; paired topology-on and topology-off conditions share the same label reassignment.

Table 5. Confirmatory cells and scientific roles.

Cell	Label mode	Topology	Role
S0	semantic	on	usefulness reference
S1	typed opaque	on	usefulness and holder control
P1–P3	full-surface permutation	on	primary ownership test
S2	semantic	off	usefulness replication reference
S3	typed opaque	off	replication holder control
P4–P6	full-surface permutation	off	ownership replication

Every condition covers the complete candidate registry. For each full-surface map, the validator checks type preservation, preservation of the within-type natural-label multiset, candidate-level bijectivity, and unchanged canonical IDs and order. For types admitting a complete value-level derangement, every candidate receives a display label different from its original label. Other types use identity carry, retaining their original labels while remaining fully visible. The map records both source-to-destination and destination-to-source provenance.

A.2 Prompt and output contract

The neutral instruction asks the model to diagnose an incident using the supplied context and evidence, select one candidate ID, and return UNKNOWN if evidence is insufficient. Both providers receive the same system and user messages and the same conceptual contract:

```
{"root_entity_id": "E007 or UNKNOWN",
  "confidence": 0.0,
  "supporting_evidence_ids": ["O001", "O002"],
  "brief_justification": "one to three concise sentences"}
```

Qwen uses provider-native JSON Schema enforcement; DeepSeek uses provider-native JSON-object mode. The local parser and validator are shared. Thinking is disabled explicitly for both provider adapters. The output cap is 1024 tokens.

A.3 Output validity details

The primary scientific value is the selected root ID. The analysis therefore records root-prediction validity and supporting-evidence validity separately. An explicit candidate ID or UNKNOWN can remain prediction-valid when evidence references exceed the maximum cardinality or fall outside the supplied evidence domain. Such auxiliary violations remain recorded.

Table 6. Confirmatory output validity.

Model	Total	Primary valid	Primary invalid	Evidence valid	Auxiliary invalid	Unknown
Qwen	870	863	7	862	8	4
DeepSeek	870	870	0	841	29	12

All seven Qwen prediction-invalid outputs ended with a provider-reported length finish reason. They remain the primary outcomes and enter the analysis unchanged.

B Seed-Level Confirmatory Results

Table 7 expands the typed-holder counts. The difference is the permuted holder count minus the seed-matched opaque holder count over 87 incidents. Aggregate intervals in the main text resample incidents after keeping the three seed rows clustered.

Table 7. Seed-level typed-holder attraction. Each seed cell reports permuted count / opaque count / percentage-point difference.

Topology	Model	20260901	20260902	20260903
On	Qwen	11/0/12.64	13/0/14.94	21/1/22.99
On	DeepSeek	2/0/2.30	2/0/2.30	3/1/2.30
Off	Qwen	8/0/9.20	10/0/11.49	15/1/16.09
Off	DeepSeek	2/0/2.30	1/0/1.15	3/1/2.30

The positive seed-level direction is consistent for both models. The prespecified confirmation rule nevertheless yields different outcomes across configurations: Qwen meets it, while DeepSeek's zero strict switches leave it unmet.

C Measurement-Sensitivity Study Details

The historical registered endpoint has attraction contrasts of 8.43 and 5.75 percentage points for Qwen and 1.92 and 1.53 points for DeepSeek under topology on and off. Registered strict-switch counts are 5, 2, 2, and 1, respectively. These values belong to the auxiliary measurement-sensitivity study and are distinct from the confirmatory endpoint.

The root-resolution audit finds 25 of 87 historical targets uniquely identical to the confirmatory roots and 62 ambiguous historical targets. All confirmatory roots resolve uniquely, and each is contained in its corresponding historical resolution set. The candidate audit covers 522 case–seed–topology rows; membership differs in every row, while omissions range from 6 to 1,720 candidates. The holder audit covers 261 case–seed rows: 3 display-string holders match the typed-provenance holder, 258 differ, and 187 display-string holders cross root type.

Typed holders remain observable in all historical treatment surfaces, permitting a complete same-response evaluation. Under typed-provenance semantics, Qwen's historical subset-surface contrast is 0.77 percentage points in both topology contexts, with confidence interval [0.00,1.92]; DeepSeek's contrast is zero in both. Because this analysis retains the historical candidate subset and changes root and holder semantics together, its estimand is evaluator-definition sensitivity. Isolating an ownership effect or decomposing the repair would require separate variation of candidate surface, root resolution, and holder definition.

Table 8. Historical same-response measurement sensitivity in percentage points. Typed-provenance evaluation retains the historically submitted candidate subsets.

Model	Top.	Registered Δ_{attr}	Typed Δ_{attr}	Typed 95% CI	Typed strict
Qwen	on	8.43	0.77	[0.00,1.92]	1
Qwen	off	5.75	0.77	[0.00,1.92]	0
DeepSeek	on	1.92	0.00	[0.00,0.00]	0
DeepSeek	off	1.53	0.00	[0.00,0.00]	0

D Reproducibility Checklist

The replication package records:

- the RCA100 version, deterministic split, bounded-view hash, candidate/evidence counts, condition maps, and their invariant and provenance checks;
- exact prompt, configuration, request, and submitted-payload hashes, together with raw and parsed outcomes, finish reasons, response metadata, and separate root/evidence validity fields;
- restricted source-resolved roots and typed holder mappings, isolated from public generation assets;
- analysis code for paired accuracy, case-clustered holder attraction, strict switches, McNemar tests, and 10,000-resample intervals; the descriptive supplement; and hashes for the measurement-sensitivity assets.

All confirmatory cases and original primary responses enter the reported results. Scoring uses exact parser outputs and fixed assignment seeds. The bootstrap seed is 20260905, and the inferential unit is the base incident.